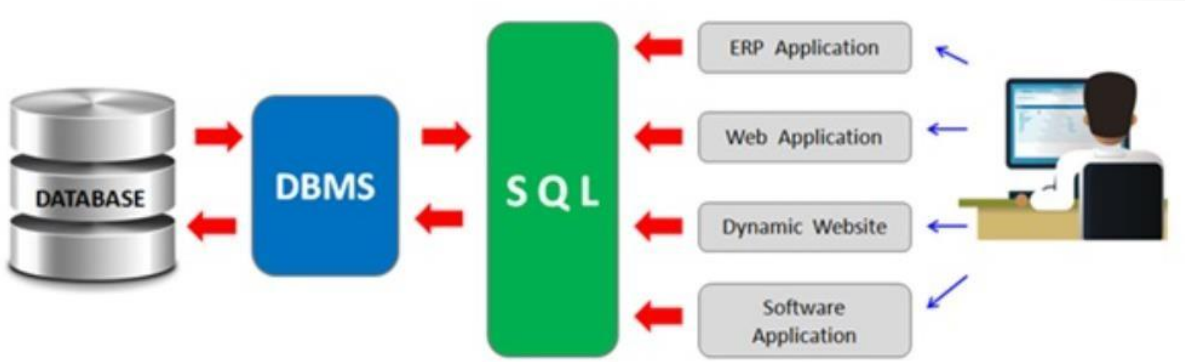
# Reference Books

- **Text Book:**
- Ramez Elmasri, Fundamentals of Database Systems (7th Edition)
- Link for Database books:
- http://auhd.edu.ye/upfiles/elibrary/Azal2020-01-22-12-28-11-76901.pdf

# LECTURE 1

# Database

- **Data:**
- Known facts that can be recorded and that have implicit meaning.
- Names, phone numbers, and addresses of the people you know.
- **Database**
- Collection of related data
- is designed, built, and populated with data for a specific purpose. It has an intended group of users and some preconceived applications in which these users are interested.
- Examples: Amazon's Database, Netflix, Facebook's database




# More examples

Emergence of web interfaces provided direct database interaction. Data on website is directly fetched from the database.

**Examples:**

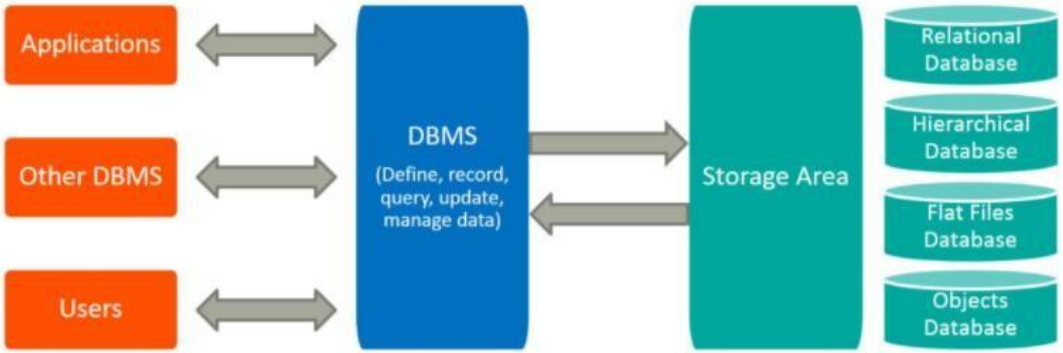
- **Banking:**

- For customer information, accounts, loans and banking transactions etc.
- **Airlines:**
- For reservations and schedule information.
- **Universities:**
- For student information, course registration and grades.

# DBMS

- **DBMS**
- Collection of interrelated data and a set of programs to access those data.
- **Goal of DBMS**

- To provide a way to store and retrieve database information that is both convenient and efficient.

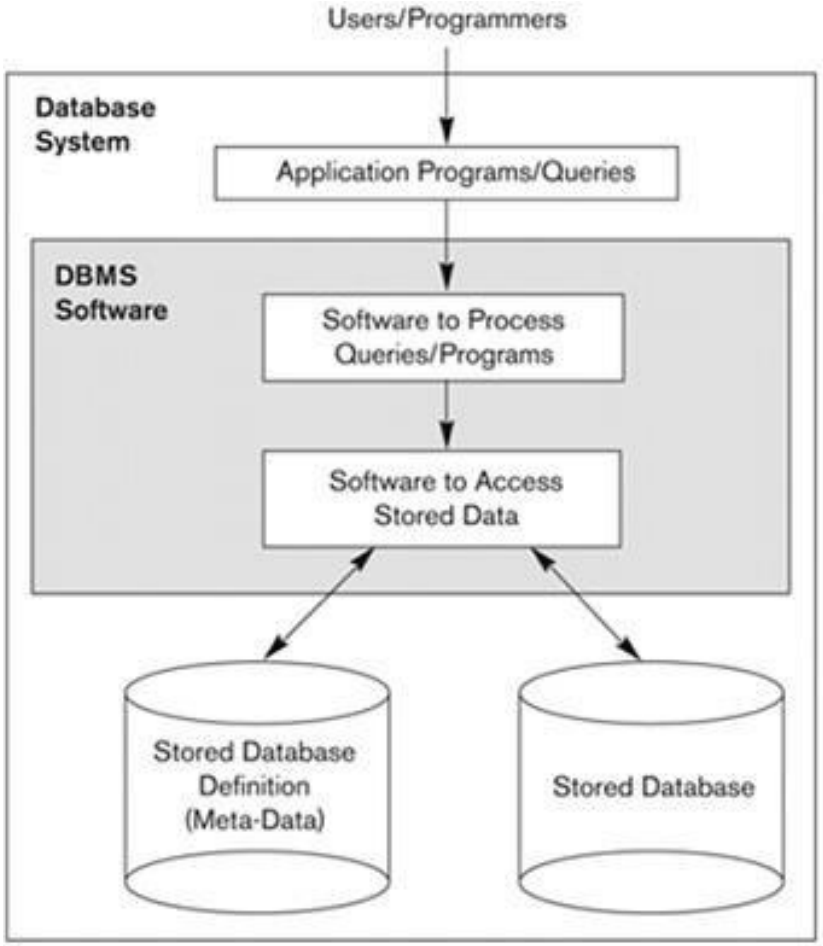



# DBMS Functionality

- **Define** a database in terms of data types, structures and constraints
- **Construct** or Load the Database on a secondary storage medium

- **Manipulating** the database by querying, generating reports, insertions, deletions and modifications to its content
- **Sharing** by a set of users and programs but keeping the data consistent at the same time.
- **Security** measures to prevent unauthorized access.
- **Presentation** and Visualization of data

# A simplified database system environment


Users/Programmers
Database System
Application Programs/Queries
DBMS Software
Software to Process Queries/Programs
Software to Access Stored Data
Stored Database Definition (Meta-Data)
Stored Database

# Some points to ponder

- What we have been using till now?
- Why are we moving from traditional file system to database system?

# ISSUES WITH FILE SYSTEMS

## Activity Time

- Lets see an excel sheet holding some records.
- Please see if you find out any problems in the given excel sheet.
- Problems related to :
- The current state of the file.
- If you want to add some other rows.
- If you want to delete some existing rows.
- If you want to update some information in the existing rows.

University - Microsoft Excel non-commercial use

A1 | Student Id

| | A | B | C | D | E | F | G | H | I | J | K | L | M |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | *Student Id* | *Student FirstName* | *Student LastName* | *Student Address* | *Student ContactNo* | *Course Id* | *Course Name* | *Course Grades* | *Instructor Id* | *Instructor Name* | *Instructor Address* | *Instructor ContactNo* | |
| 2 | S1 | Jerry | Robinson | Obere Str. 0123 | 030-3456789 | UV101 | Linear Algebra | A | I_01 | Paul Koch | 9012, rue Royale | 34.56.78.90 | |
| 3 | S1 | Jerry | Robinson | Obere Str. 0123 | 030-3456789 | UV204 | Visual Basic | B | I_03 | Michael Allen | Kirchgasse 9012 | (171) 789-0123 | |
| 4 | S1 | Jerry | Robinson | Obere Str. 0123 | 030-3456789 | UV301 | Wellness | A | I_02 | Mark Hassall | Fauntleroy Circus 4567 | (1) 890-1234 | |
| 5 | S1 | Jerry | Robinson | Obere Str. 0123 | 030-3456789 | UV404 | Political Science | C | I_10 | John Peoples | Cerrito 3456 | (5) 456-7890 | |
| 6 | S1 | Jerry | Robinson | Obere Str. 0123 | 030-3456789 | UV501 | Network Security | A | I_13 | Tom Higgin | Hauptstr. 0123 | (11) 012-3456 | |
| 7 | S2 | John | Walker | Avda. de la 5678 | (5) 789-0123 | UV110 | Calculus-2 | C | I_41 | Venky Krishnan | 6789 Baker Blvd. | (503) 555-0122 | |
| 8 | S2 | John | Walker | Avda. de la 5678 | (5) 789-0123 | UV208 | MicroComputer | B | I_23 | Shabalin Rostisl | C/ Romero, 1234 | (95) 901 23 45 | |
| 9 | S2 | John | Walker | Avda. de la 5678 | (5) 789-0123 | UV303 | Calculus | C | I_12 | Jacek Jelitto | Hauptstr. 0123 | 0452-678901 | |
| 10 | S2 | John | Walker | Avda. de la 5678 | (5) 789-0123 | UV404 | Political Science | A | I_10 | John Peoples | Cerrito 3456 | (5) 456-7890 | |
| 11 | S2 | John | Walker | Avda. de la 5678 | (5) 789-0123 | UV501 | Network Security | A | I_13 | Tom Higgin | Hauptstr. 0123 | (11) 012-3456 | |
| 12 | S3 | Mary | Mosis | Mataderos 7890 | (5) 123-4567 | UV101 | Linear Algebra | A | I_01 | Koch, Paul | 9012, rue Royale | 34.56.78.92 | |
| 13 | S3 | Mary | Mosis | Mataderos 7890 | (5) 123-4567 | UV204 | Visual Basic | B | I_03 | Michael Allen | Kirchgasse 9012 | (171) 789-0123 | |
| 14 | S3 | Mary | Mosis | Mataderos 7890 | (5) 123-4567 | UV301 | Wellness | A | I_02 | Mark Hassall | Fauntleroy Circus 4567 | (1) 890-1234 | |
| 15 | S3 | Mary | Mosis | Mataderos 7890 | (5) 123-4567 | UV407 | Geography | B | I_20 | Tianna Jones | Walserweg 4567 | (604) 901-2345 | |

# 1. Redundancy

- Duplicates of data
- Same piece of information present at multiple locations

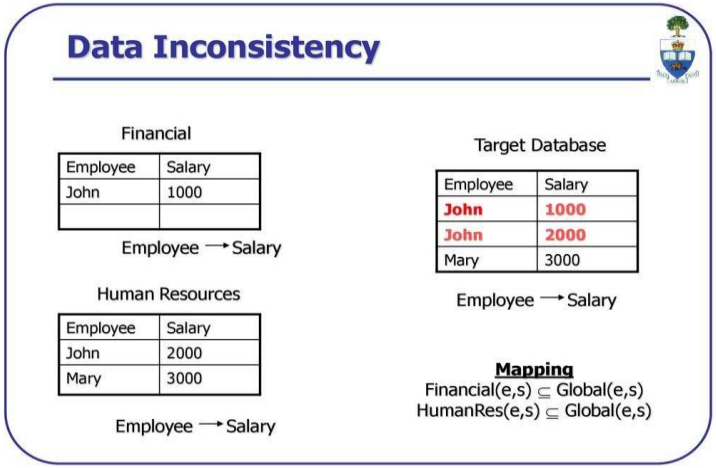
| Patient Id | Name | D.o.B | Gender | Phone | Doctor Id | Doctor | Room |
|---|---|---|---|---|---|---|---|
| 134 | Jeff | 4-Jul-1993 | Male | 7876453 | 01 | Dr Hyde | 03 |
| 178 | David | 8-Feb-1987 | Male | 8635467 | 02 | Dr Jekyll | 06 |
| 198 | Lisa | 18-Dec-1979 | Female | 7498735 | 01 | Dr Hyde | 03 |
| 210 | Frank | 29-Apr-1983 | Male | 7943521 | 01 | Dr Hyde | 03 |
| 258 | Rachel | 8-Feb-1987 | Female | 8367242 | 02 | Dr Jekyll | 06 |

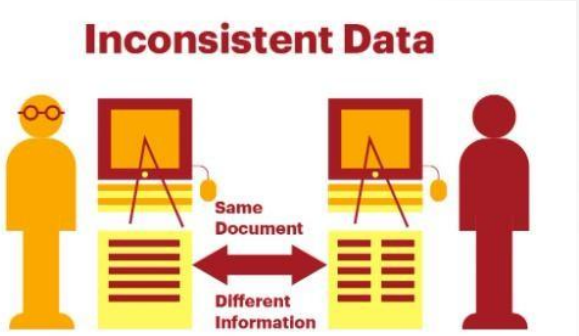
Duplicate

Duplicate

# 2. Inconsistency

- Different information related to same entity in different/same files.
- Data can be inconsistent if:
- Changes are made to one place and not in another
- There is a typo
- Data is present in different formats
- Data redundancy can cause data inconsistency




# 3. Data Isolation & Data Mapping

- Data is scattered in various files and may be in different formats.
- Writing new programs for different files may be difficult.
- No link between the files
- Changes in one cannot be reflected in another

| IID | LNAME | FNAME | RANK | SALARY | DEPT |
|---|---|---|---|---|---|
| 201 | Gauss | Fred | Prof | 55000 | Math |
| 301 | Joyce | James | Assoc Prof | 23000 | English |

| CID | TITLE | SEMESTER | UNITS | LOCATION | TIME | INSTRUCTOR |
|---|---|---|---|---|---|---|
| MATH23 | Calculus | Fall 2000 | 3 | MSRI1 | MWF 1-2 | 201 |
| ENG1A | Literature | Fall 2000 | 5 | DWIN2 | TTh 3-4 | 301 |
| | | | | | | |

# 4. Integrity Problems

- Age must be under 60
- Account balance must not be negative
- Salary must be less than 10,000. If this rule changes, changes would be done at every check.
- How to handle all these?

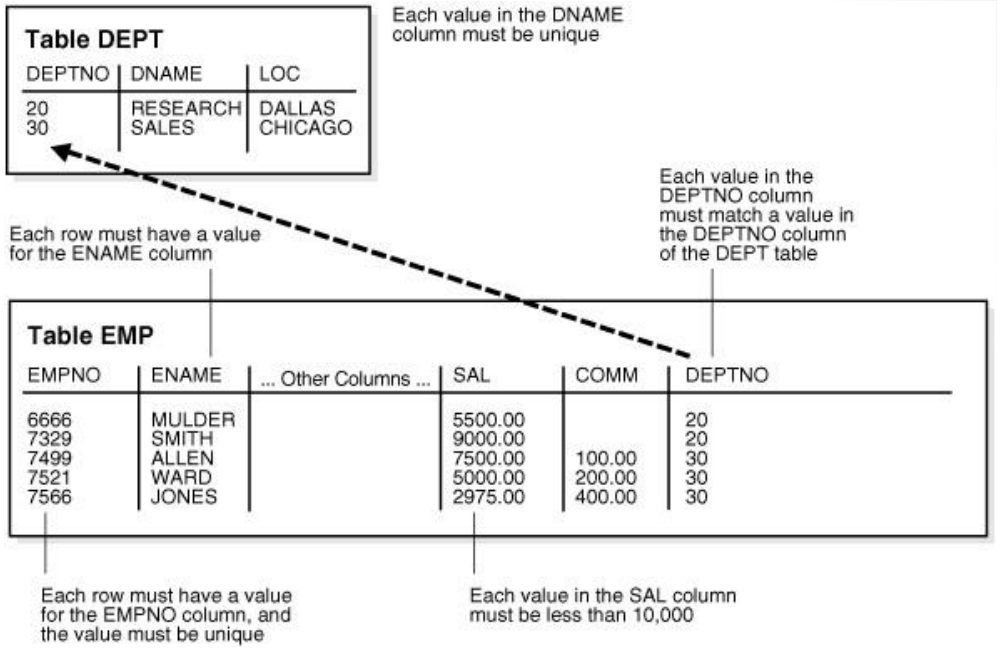


Table DEPT
DEPTNO DNAME LOC
20 RESEARCH DALLAS
30 SALES CHICAGO
Each value in the DNAME column must be unique
Each value in the DEPTNO column must match a value in the DEPTNO column of the DEPT table
Each row must have a value for the ENAME column
Table EMP
EMPNO ENAME ... Other Columns ... SAL COMM DEPTNO
6666 MULDER 5500.00 20
7329 SMITH 9000.00 20
7499 ALLEN 7500.00 100.00 30
7521 WARD 5000.00 200.00 30
7566 JONES 2975.00 400.00 30
Each row must have a value for the EMPNO column, and the value must be unique
Each value in the SAL column must be less than 10,000

# 5. Atomicity

# *Non-Atomic Data*

These attributes contain non-atomic data (also sometimes called ***repeating data***);

*Example* : a "relation" containing non-atomic data.

| Part | Component | Quantity |
|---|---|---|
| Bicycle | Frame | 1 |
| | Wheel | 2 |
| Frame | A-frame | 1 |
| | Handlebars | 1 |
| | Saddle | 1 |
| Wheel | Rim | 1 |
| | Axle | 1 |
| | Spoke | 40 |

but this attribute does contain atomic data.

# Some other issues

- Ad hoc Reporting
- created for one-time use to find the answer to a specific business questions.
- i.e., how many support tickets were resolved last week or how many calls a salesperson made yesterday.
- Data Security
- grant and revoke privileges, role-based access control
- User Interfaces
- Because many types of users with varying levels of technical knowledge use a database, a DBMS should provide a variety of user interfaces.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- Self describing nature
- use of a catalog to store the database description (schema) to make it self-describing

# Metadata/Data Dictionary/Catalog

| emlployee_id | first_name | last_name | nin | department_id |
|---|---|---|---|---|
| 44 | Simon | Martinez | HH 45 09 73 D | 1 |
| 45 | Thomas | Goldstein | SA 75 35 42 B | 2 |
| 46 | Eugene | Cornelsen | NE 22 63 82 | 2 |
| 47 | Andrew | Petculescu | XY 29 87 61 A | 1 |
| 48 | Ruth | Stadick | MA 12 89 36 A | 15 |
| 49 | Barry | Scardelis | AT 20 73 18 | 2 |
| 50 | Sidney | Hunter | HW 12 94 21 C | 6 |
| 51 | Jeffrey | Evans | LX 13 26 39 B | 6 |
| 52 | Doris | Berndt | YA 49 88 11 A | 3 |
| 53 | Diane | Eaton | BE 08 74 68 A | 1 |
| 54 | Bonnie | Hall | WW 53 77 68 A | 15 |
| 55 | Taylor | Li | ZE 55 22 80 B | 1 |

Data

Metadata

| Column | Data Type | Description |
|---|---|---|
| emlployee_id | int | Primary key of a table |
| first_name | nvarchar(50) | Employee first name |
| last_name | nvarchar(50) | Employee last name |
| nin | nvarchar(15) | National Identification Number |
| position | nvarchar(50) | Current postion title, e.g. Secretary |
| department_id | int | Employee deparmtnet. Ref: Departmetns |
| gender | char(1) | M = Male, F = Female, Null = unknown |
| employment_start_date | date | Start date of employment in organization. |
| employment_end_date | date | Employment end date. Null if employee st |

# DESIGN PHASES OF DATABASES

## How is a house constructed?

- If you want to build a house what steps will you follow?
- We will give our requirements to the architect.
- He will show us some paperwork i.e. a map or a 3d model.
- Then the foundations are laid and grey structure is built according to the preferred material.
- We then finalize about the finishing and furnishing.

# Design Phases of Database

1. Requirement Definition and analysis

- Design of a new application for an existing database

OR

- Design of a new database

2. Conceptual Design

-These requirements are documented in detail to form a conceptual design

-Conceptual design can be represented and manipulated using some computerized tool so that it can easily be maintained, modified and transformed to database implementation.

Example: Entity Relationship Model

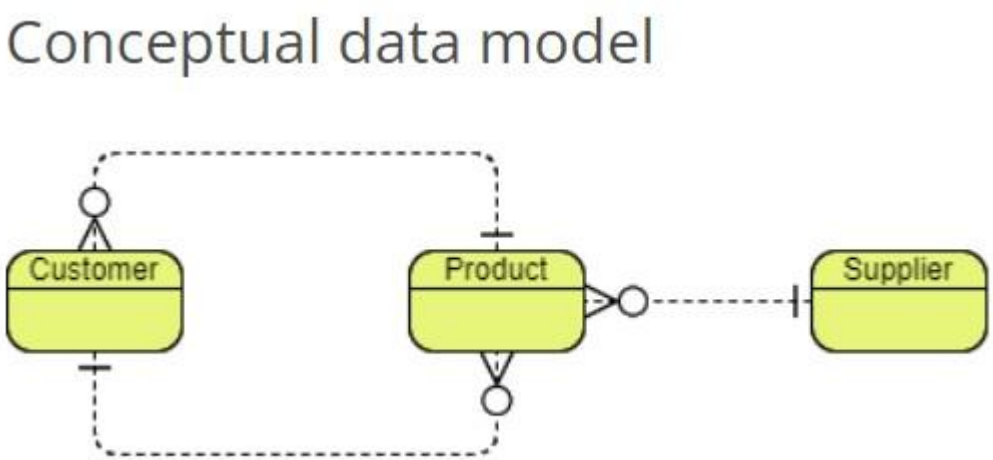
# Design Phases of Database

3. Logical Design
   - Conceptual design is transformed into logical design
   - Logical design can be expressed in a **data model**
   - Example: Relational Data Model

4. Physical Design
   - In this stage further specifications are provided for storing and accessing the database.

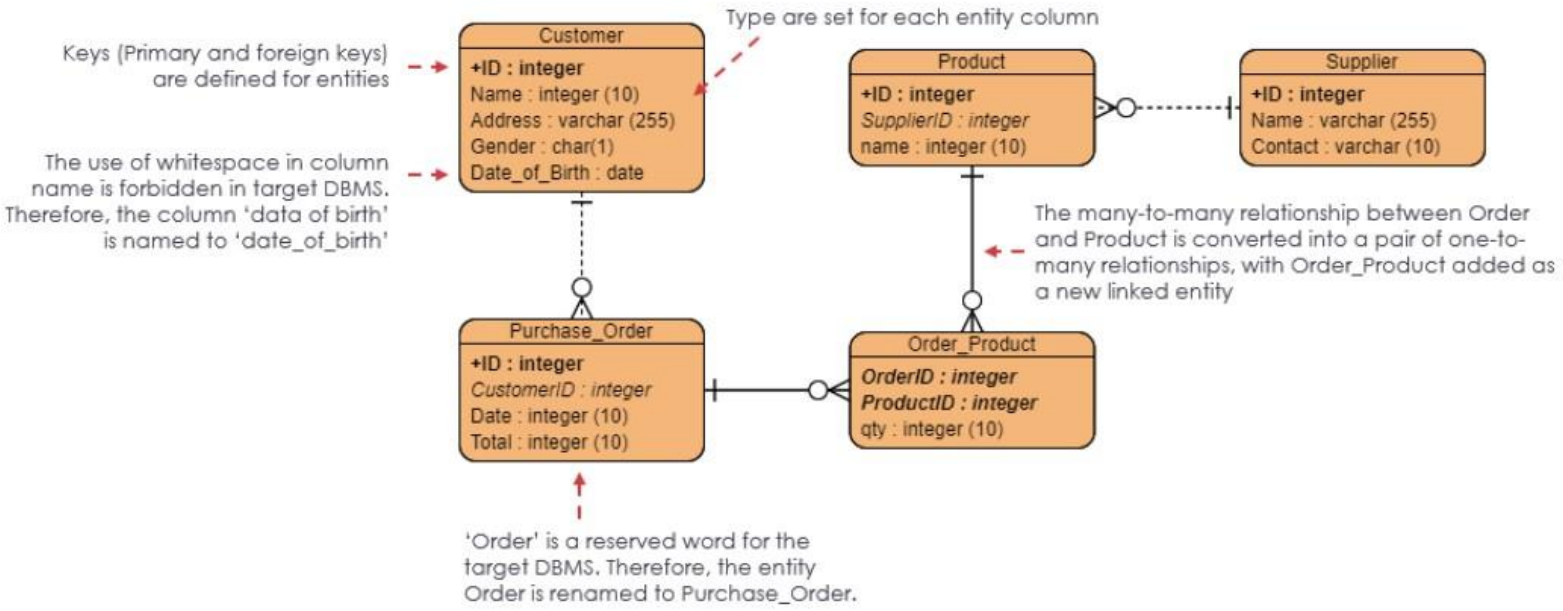
- Database design is implemented , populated with actual data and continuously maintained to reflect the actual state of miniworld.
- Click here

# Conceptual data model – ER Diagram



# Physical Model

Physical data model example



Users of Database system

When not to use a db?

# SELF STUDY TOPICS

# Database Users

**Actors on the scene**

- **Database administrators:** responsible for authorizing access to the database, for coordinating and monitoring its use, acquiring software, and hardware resources, controlling its use and monitoring efficiency of operations.
- **Database Designers:** responsible to define the content, the structure, the constraints, and functions or transactions against the database. They must communicate with the end-users and understand their needs.
- **End-users:** they use the data for queries, reports and some of them update the database content.

# Categories of End-users

- **Casual** : access database occasionally when needed
- **Naïve or Parametric** : they use previously well-defined queries knows as“canned transactions”against the database.
- They don't have any DBMS knowledge but use the DB applications to get the desired results
- Examples are bank tellers/clerks or reservation clerks who do this activity for an entire shift of operations.
- Railway's ticket booking users

# Categories of End-users

- **Sophisticated** : these include business analysts, scientists, engineers, others thoroughly familiar with the system capabilities. Many use tools in the form of software packages that work closely with the stored database.
- **Stand-alone** : mostly maintain personal databases using ready-to-use packaged applications. An example is a tax package that basically stores a variety of personal financial data for tax purposes.

# Categories of End-users

- **Application Programmers/ Software Engineers**
- Determine the requirements of end users, especially naive and parametric end users, and writes application programs that uses the database.
- Programming languages used include COBOL, Java etc.
- Application programs meet the user requirements such as retrieving, creating and updating information
- **DBMS system designers and implementers**

- Design and implement the DBMS modules and interfaces as a software package.

# Workers Behind the Scene

- **DBMS system designers and implementers**
- Design and implement the DBMS modules and interfaces as a software package.

- **Operators and maintenance personnel**
- **They** are responsible for the actual running and maintenance of the hardware and software environment for the database system.

# When not to use a DBMS

- **Main inhibitors (costs) of using a DBMS**:
  - High initial investment and possible need for additional hardware.
- **When a DBMS may be unnecessary:**
  - If the database and applications are simple, well defined, and not expected to change.
  - If access to data by multiple users is not required.

# Thank You.